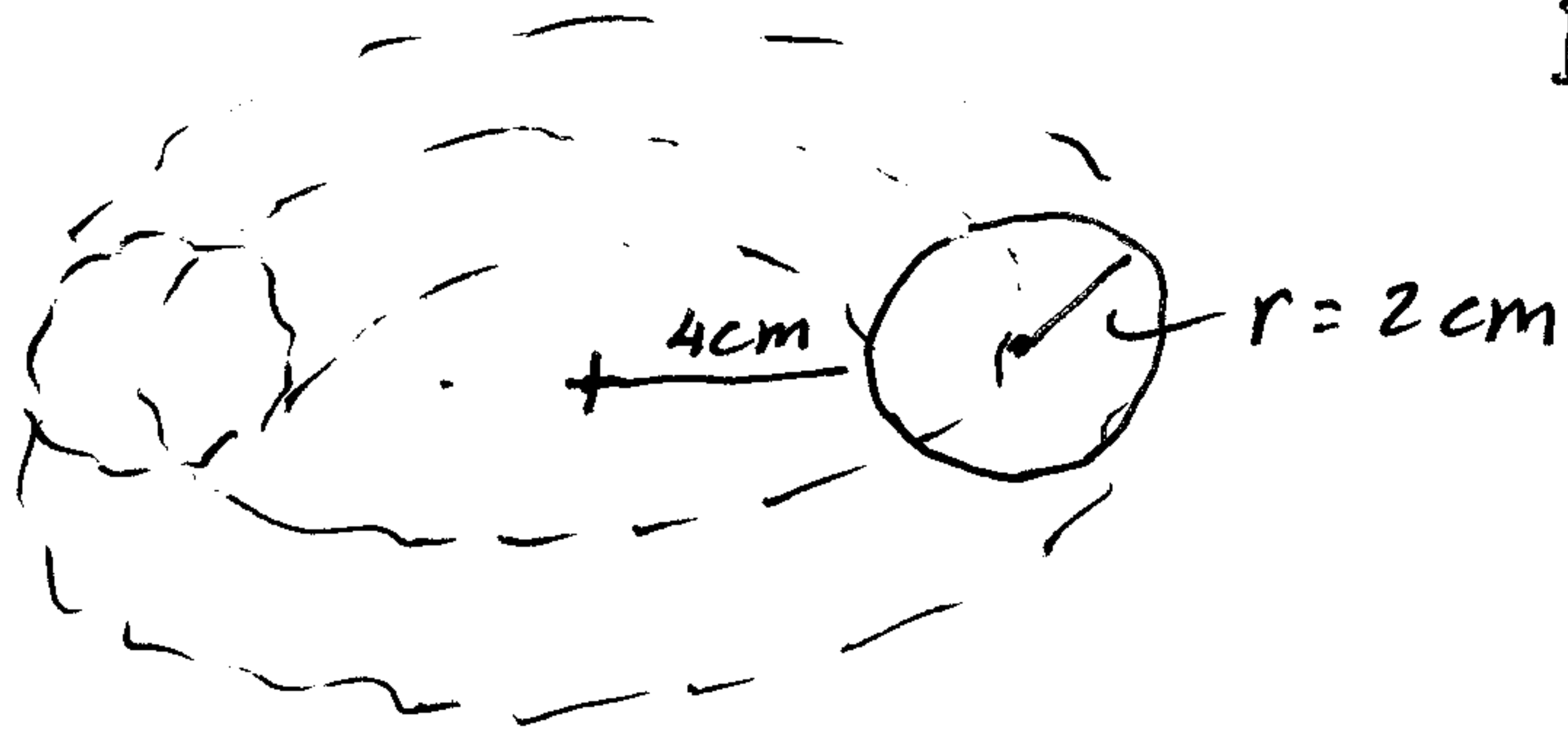


Problem a few years ago:



Donut Shape ...
(Toroid)

$$Vol = \sum \theta \bar{r} A$$

$$\begin{aligned} Vol &= 2\pi (4+2)(\pi \cdot 2^2) \\ &= 2\pi (6(4\pi)) \\ &= 48\pi^2 \end{aligned}$$

$$\boxed{V = 473.7 \text{ cm}^3}$$

$$A = \sum \theta \bar{r} L$$

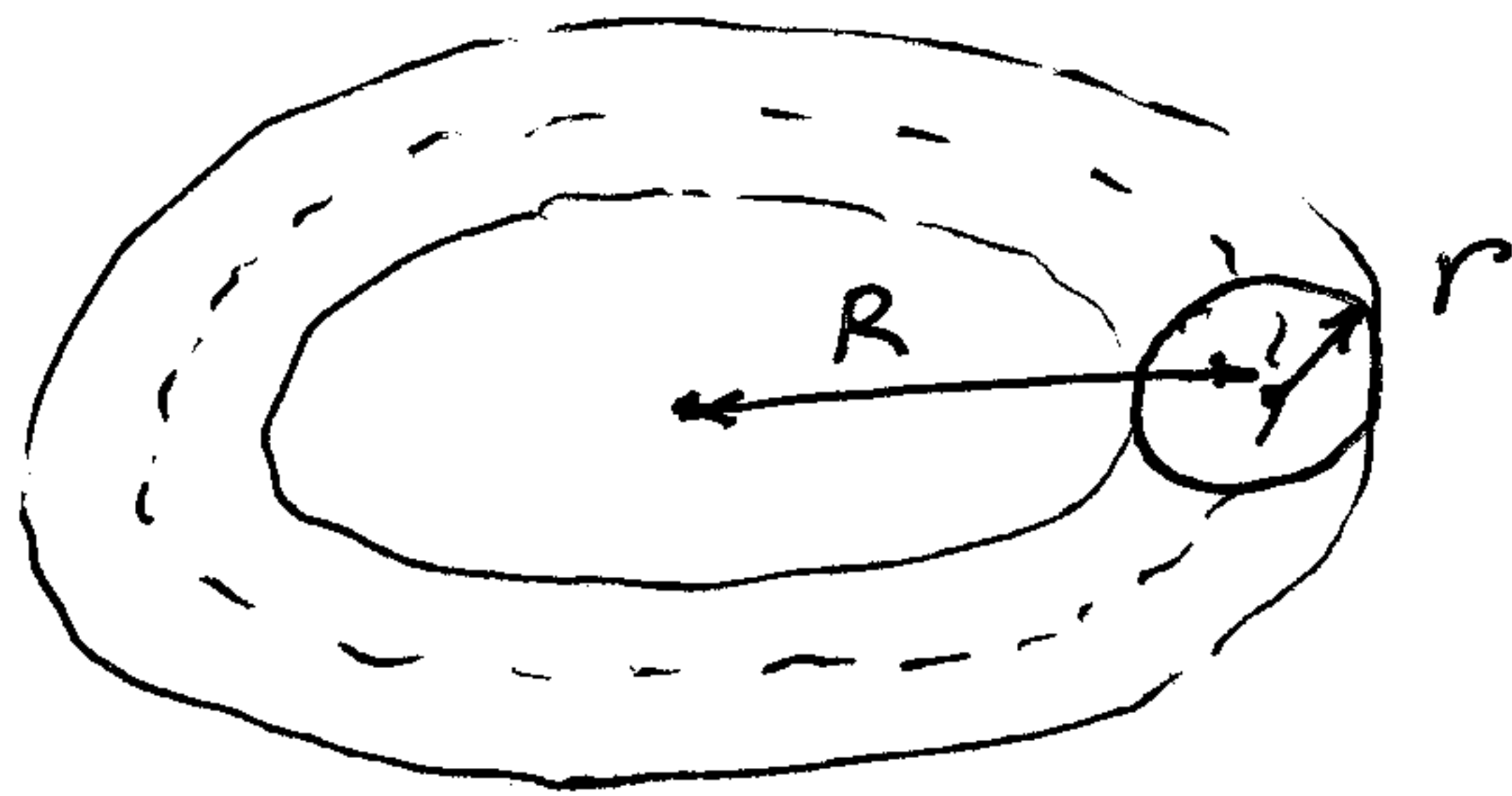
$$\begin{aligned} A &= 2\pi (4+2)(2\pi (2)) \\ &= 2\pi (6)(4\pi) \\ &= 48\pi^2 \end{aligned}$$

$$\boxed{A = 473.7 \text{ cm}^2}$$

Torus (Donut)

$$A = 4\pi^2 R r$$

$$V = 2\pi^2 R r^2$$



Our example: $R = 4 + 2 = 6 \text{ cm}$
 $r = 2 \text{ cm}$

$$\begin{aligned} A &= 4\pi^2 (3r)(r) = 12r^2\pi^2 = A \\ V &= 2\pi^2 (3r)(r^2) = 6\pi^2 r^3 = V \end{aligned} \quad \left. \vphantom{\begin{aligned} A &= 4\pi^2 (3r)(r) \\ V &= 2\pi^2 (3r)(r^2) \end{aligned}} \right\} \begin{array}{l} \text{If} \\ R = 3r \end{array}$$

$$A = 4\pi^2 (6)(2) = 48\pi^2$$

$$V = 2\pi^2 (6)(4) = 48\pi^2$$

and for $r = 2 \text{ cm}$, $R = 6 \text{ cm}$

$$12r^2 = 48$$

$$6r^3 = 6(8) = 48$$

$$\boxed{A = V}$$

Another case: If $R = 3r$, but $r = 3 \text{ cm}$
 $R = 9 \text{ cm}$

$$A = 12(3)^2\pi^2 = 108\pi^2$$

$$V = 6\pi^2 (3)^3 = 162\pi^2$$

$$\boxed{A < V}$$

Another case: If $R = 3r$, but $r = 1 \text{ cm}$
 $R = 3 \text{ cm}$

$$A = 4\pi^2 (3)(1) = 12\pi^2$$

$$V = 2\pi^2 (3)(1)^2 = 6\pi^2$$

$$\boxed{A > V}$$